DEVOPS CAPSTONE · SESSION 21

LiftLog

A strength-training tracker, built, secured and shipped with GitOps on Amazon EKS



Live on EKS behind an AWS Application Load Balancer

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What LiftLog does



Exercise library

21 seeded exercises by muscle group



Workout logging

Sets with weight, reps and RPE



History

Every session with volume per workout



Personal records

Best set ranked by estimated 1RM



Volume analytics

Weekly totals and muscle-group split



Progress charts

Estimated 1RM over time per lift

11

REST endpoints

14

automated tests

3

PRO analytics views

Tech stack



Frontend

React 19 · Vite 8 · nginx (non-root)



Backend

Python 3.12 · FastAPI · SQLAlchemy · Alembic



Database

PostgreSQL 16 on a gp3 EBS volume



Containers

Docker multi-stage · Compose



CI/CD

GitHub Actions · GHCR · Trivy



Infrastructure

Terraform · AWS VPC · EKS 1.36



Kubernetes

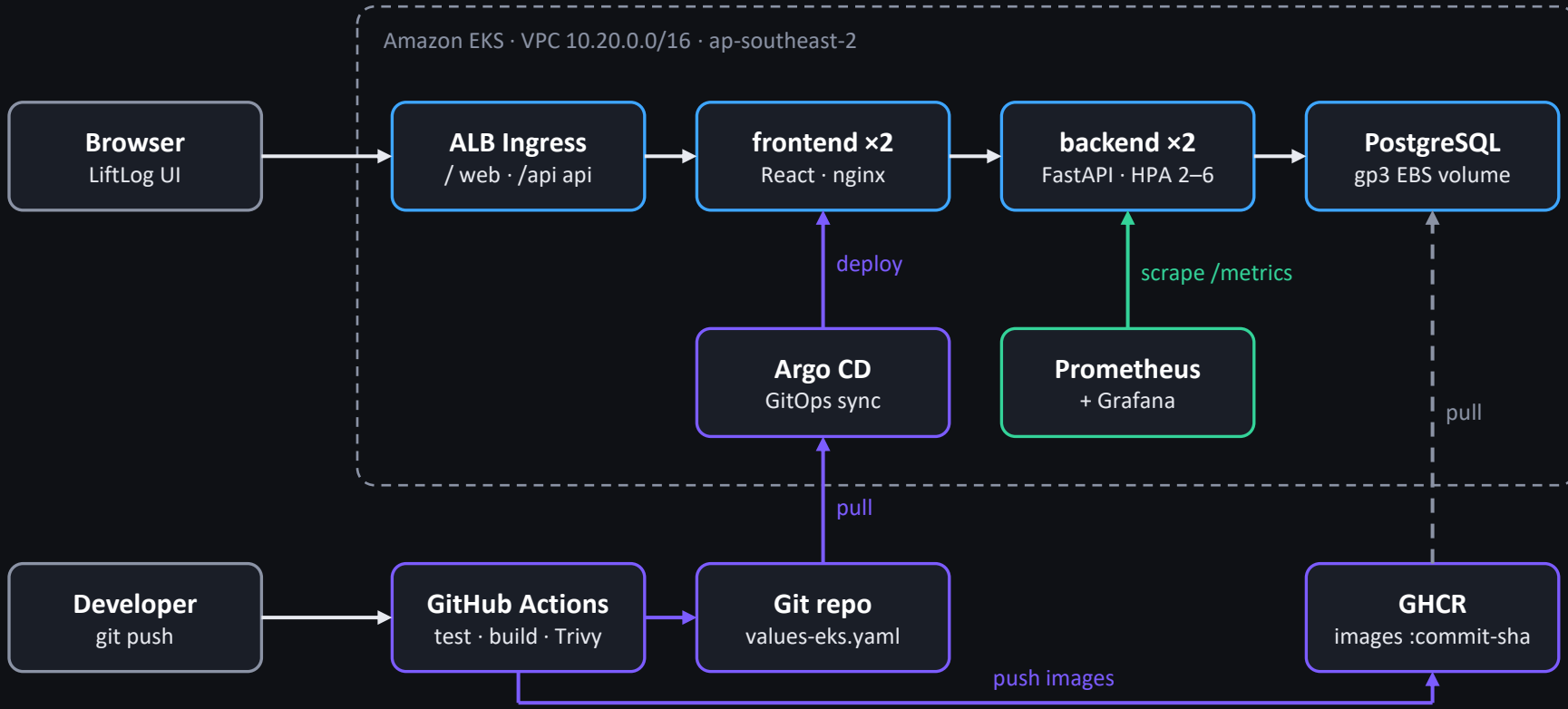
Helm · ALB Ingress · HPA · Argo CD



Observability

Prometheus · Grafana · ServiceMonitor

Architecture



CI/CD: from git push to production



1. Push	2. Test	3. Build	4. Trivy gate	5. Publish	6. GitOps	7. Argo CD
git push to main	pytest (14) + frontend build	backend + frontend images	fail on fixable HIGH/CRITICAL	GHCR, tagged :commit-sha	commit the SHA to Git	sync + rolling update

3 jobs

test → build & scan → GitOps

≈ 2 min

push to green pipeline

0

cloud credentials stored in CI

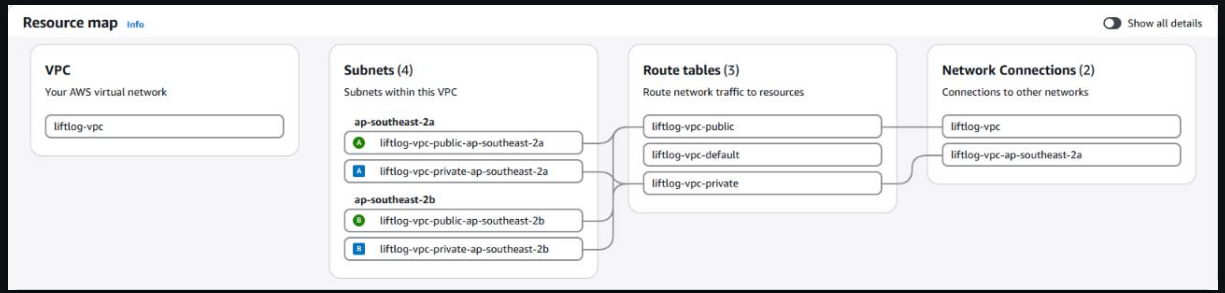
Infrastructure as code with Terraform

64
resources in one plan

2 + 2
public + private subnets

1.36
EKS Kubernetes version

2
m7i-flex.large worker nodes



liftlog-vpc in the AWS console: subnets, route tables, internet gateway

terraform plan
64 to add

terraform apply
VPC + EKS ≈ 20 min

terraform destroy
64 destroyed, $0 left

Kubernetes on EKS: Helm + Argo CD

Helm chart

backend + frontend ×2, HPA 2–6, ServiceMonitor

ALB Ingress

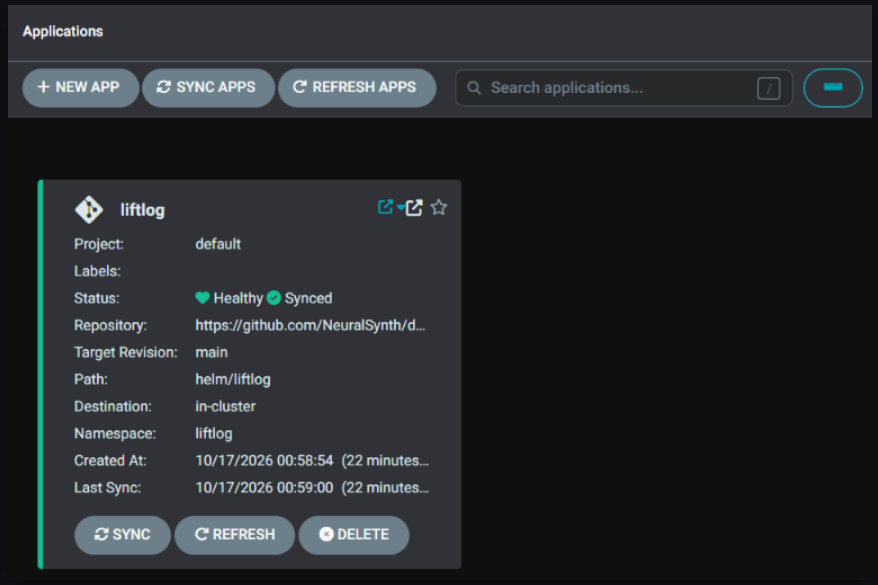
/ → frontend, /api → backend

Persistent Postgres

gp3 EBS volume via the EBS CSI driver

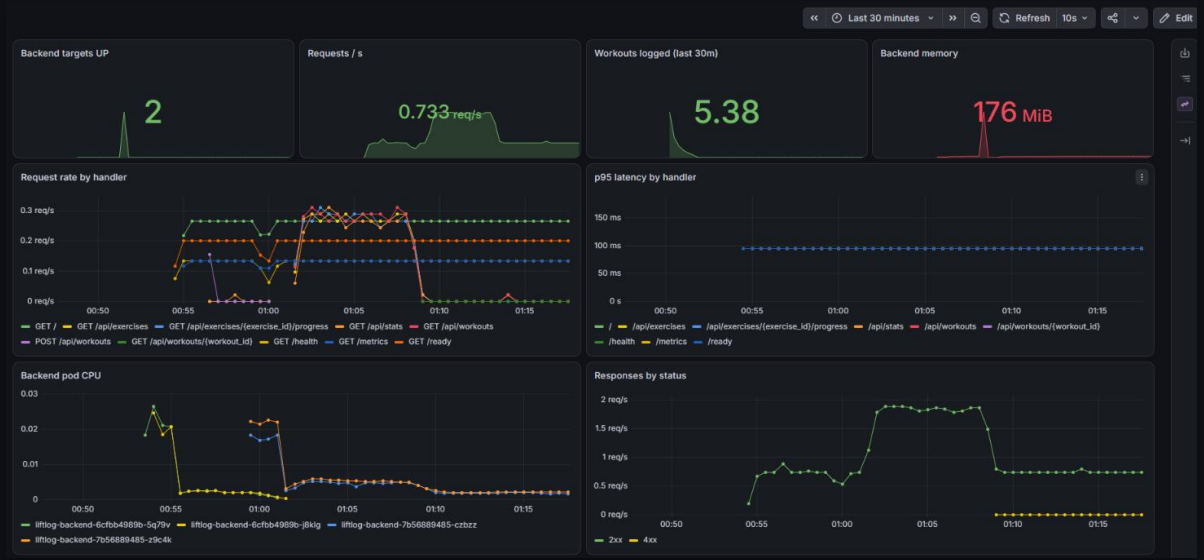
Argo CD

auto-sync and self-heal from Git



Argo CD on EKS: liftlog is Healthy and Synced to main

Observability: Prometheus + Grafana



Grafana dashboard-as-code: LiftLog API (EKS)

2/2

backend targets UP

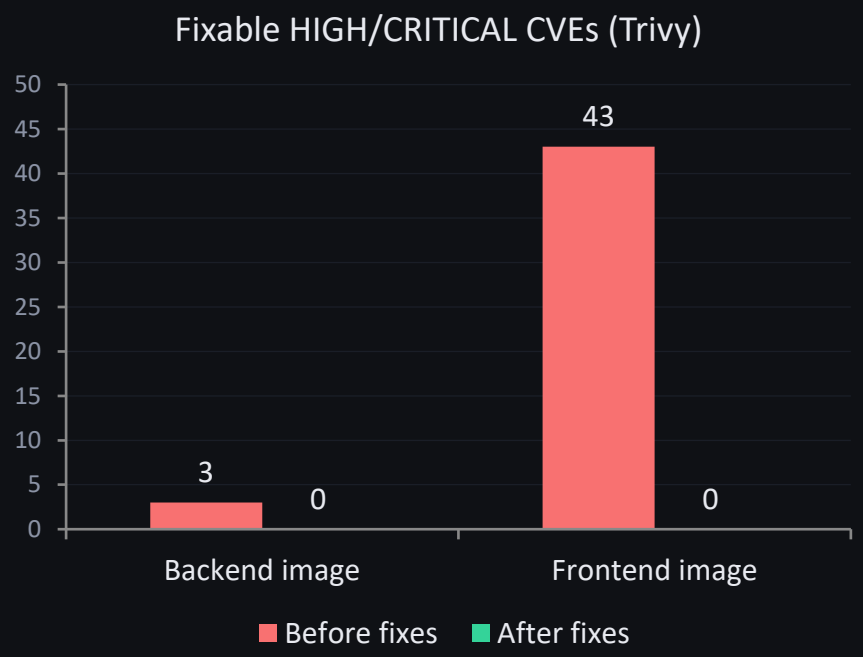
8

panels, defined in Git

≈ 95 ms

p95 API latency

DevSecOps: security built into the pipeline



Trivy gate

fixable HIGH/CRITICAL fails the build

Non-root containers

backend uid 10001, frontend uid 101

No long-lived keys

aws login, GITHUB_TOKEN, Pod Identity

Fixed, not ignored

FastAPI/starlette upgrade, new nginx base

Key learnings

Respect the guardrails

An SCP blocked the IAM OIDC provider, so I moved to EKS Pod Identity.

Versions cost money

K8s 1.31 was in extended support at 6× the price; 1.36 is standard.

Storage has sharp edges

EBS lost+found broke initdb; PGDATA moved to a subdirectory.

No creds in CI

CI writes the image SHA to Git; Argo CD pulls it.

Fix, don't ignore

46 fixable CVEs cleared by upgrades, not ignore rules.

Tear down in order

Argo CD app, then Helm (ALB, EBS), then terraform destroy.

Thank you

LiftLog: tested, scanned, shipped with GitOps, and observed on Amazon EKS



Repository

NeuralSynth/devops-heros-s21-liftlog



Walkthrough video

youtu.be/2FtC1-wvGKg



Evidence

docs/evidence in the repository